3.1 Composition of vegetables and fruits

Vegetables contain a high amount of water. They also contain carbohydrates, dietary fibre, protein, vitamins and other nutrients that are important for human health.

Lettuce, cucumbers and leafy vegetables contain about 95% water, therefore only 5% of their mass is dry matter. Hard vegetables like carrot and pumpkin have around 12-15% dry matter.

Carbohydrates are the main components of vegetables and fruit. They represent more than 90% of their dry matter. Carbohydrates are present as starch, sugars and dietary fibre. Starch is mainly found in root vegetables, such as potatoes and sweet potatoes.

The main sugars that are present in vegetables and fruits are glucose, sucrose and fructose. Although more usually associated with fruit, sugars are an important component of flavour in vegetables such as carrots, sweet corn and peas.

Dietary fibre compounds like cellulose, lignin, pectins and other substances are also found in vegetables and fruits. Dietary fibre in vegetables and fruits have several health benefits like lowering blood sugar and cholesterol levels.

3.2 Classification of Vegetables

Vegetables are classified according to the part of the plant consumed, colour of the vegetable or according to the nutritive value.

3.3 Nutritive value of vegetables

- Roots and tubers like carrots, potatoes and sweet potatoes contain a large amount of starch (carbohydrates), hence they contribute to energy value of the food.

Table 3.1 Classification of vegetables based on parts of plants consumed

Parts of plants	Examples
Roots	Carrot, beet root, radish, turnip, colocasia
Tubers	Potatoes, sweet potatoes, tapioca
Bulb	Onion, garlic, leeks
Leaves	Cabbage, lettuce, spinach, amaranth, fenugreek leaves, coriander leaves, mint leaves, greens
Flowers	Plantain flower, cauliflower, broccoli
Fruits	Tomatoes, brinjal, lady's finger, pumpkin, cucumber, gourds (ash gourd, bottle gourd), capsicum, drumstick, plantain
Legumes (pods and seeds)	Peas, beans, chowli, broad beans, French beans, double beans, Bengal gram tender, red gram tender.
Stems	Plantain stem, ginger, amaranth stem, celery stem, lotus stem and greens

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